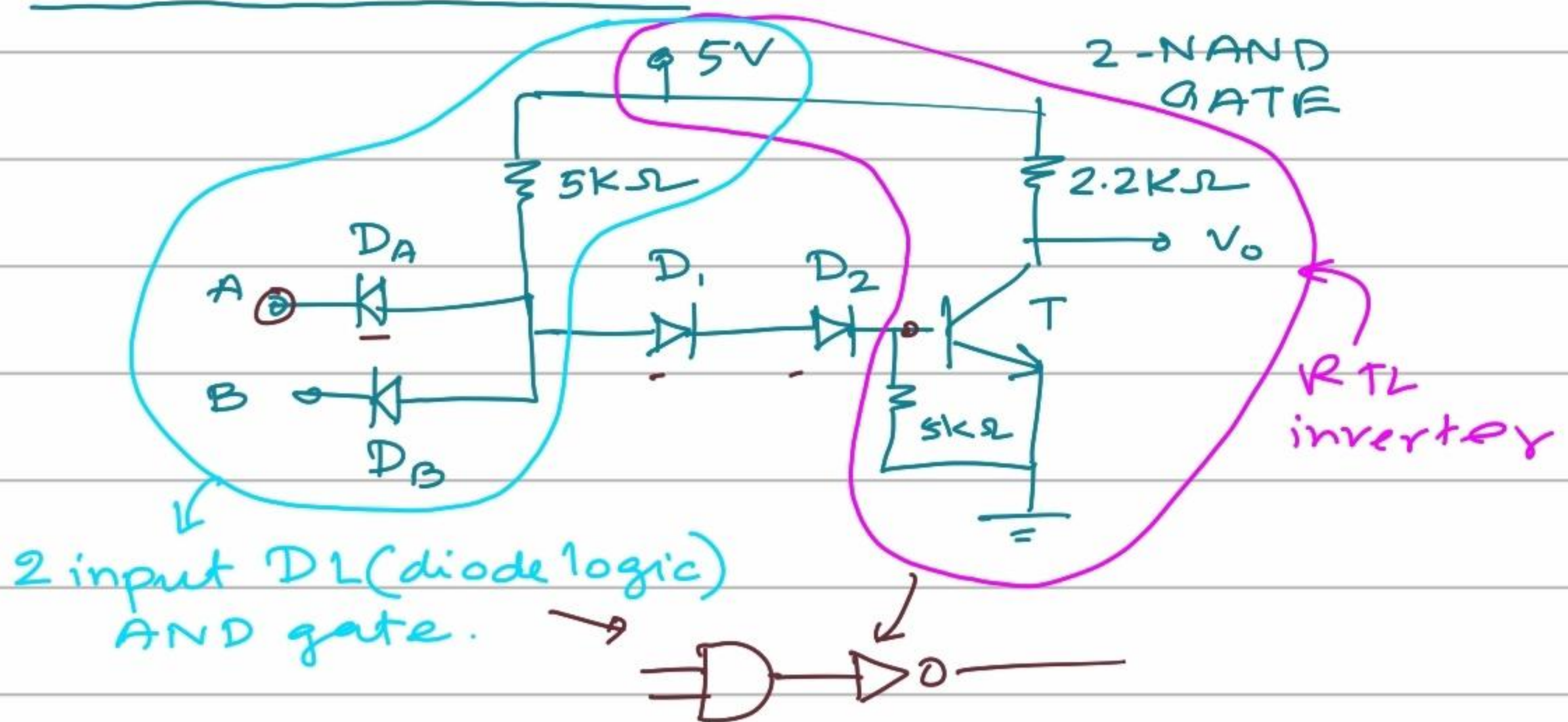


DTL (Diode Transistor Logic)

* Bipolar $\rightarrow$ Saturated
(Switching transistor will operate in saturation and cutoff mode).

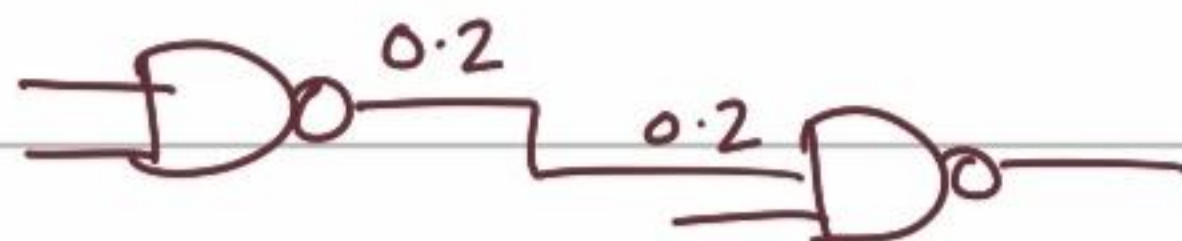
* Input terminals connects through diodes to the switching transistor's base terminal.

DTL NAND GATE:



Basic operation :

Calculate the currents and voltages of the following DTL gate and verify that it works like a NAND gate. Assume that input stages might be connected to the output stage of other DTL NAND gate.
 $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.2V$. $\beta_F = 30$.

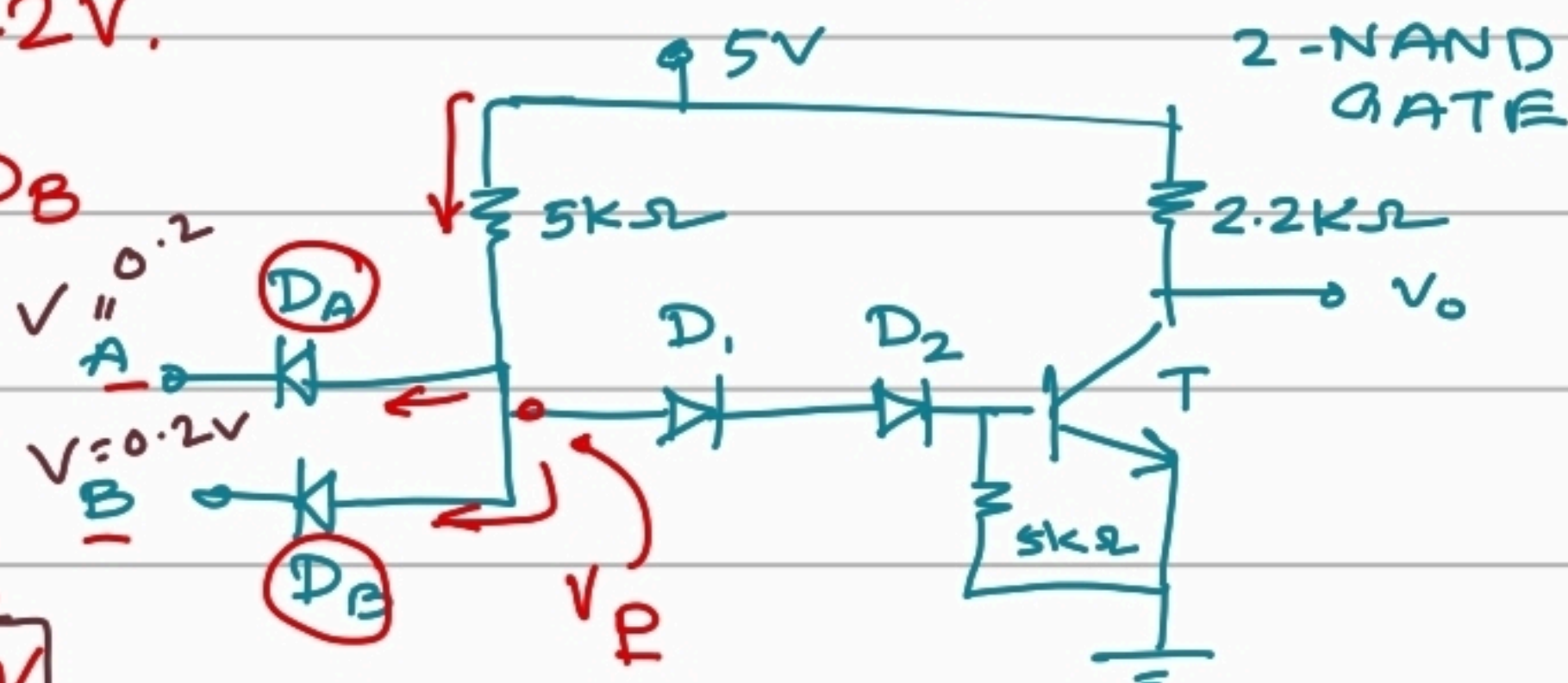
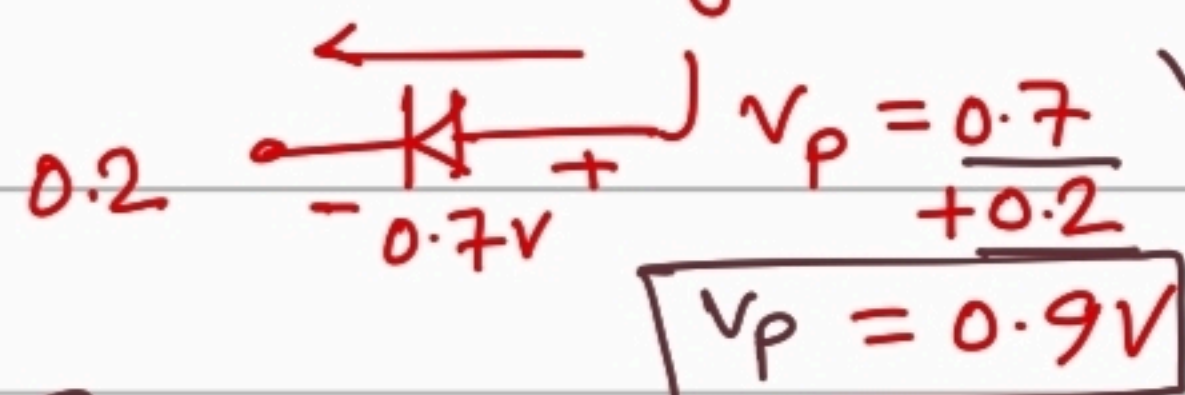


Ans:

Case ① Both inputs are low.

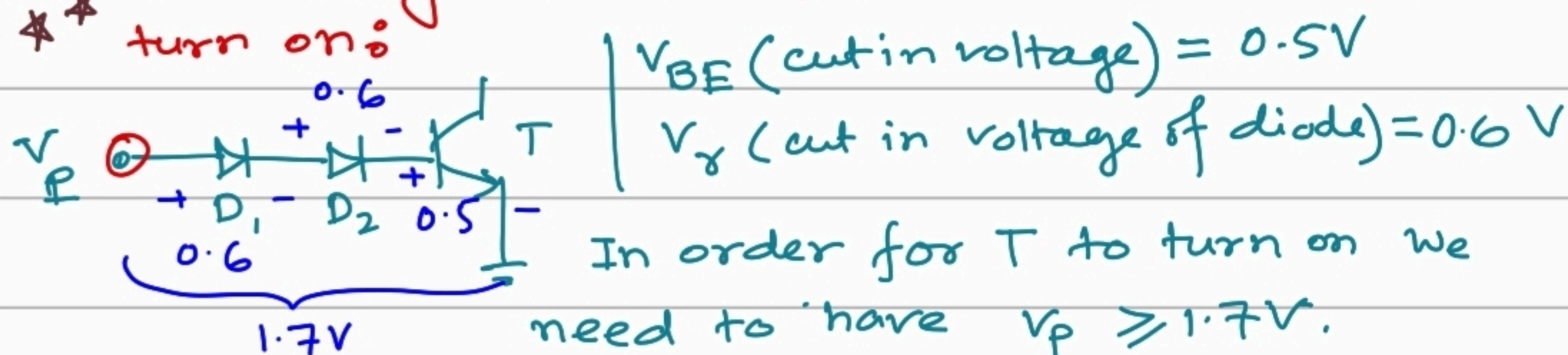
Because inputs are connected to some other output stage, the output voltage = input voltage = $0.2V$. $V_A = V_B = 0.2V$.

Assume, D_A and D_B are conducting.



Since we are assuming diodes are conducting the anode voltage V_P of D_A and D_B must be $0.7V$ higher than the cathode voltage. Thus $V_P = 0.9V$.

□ Necessary condition for transistor T to turn on:



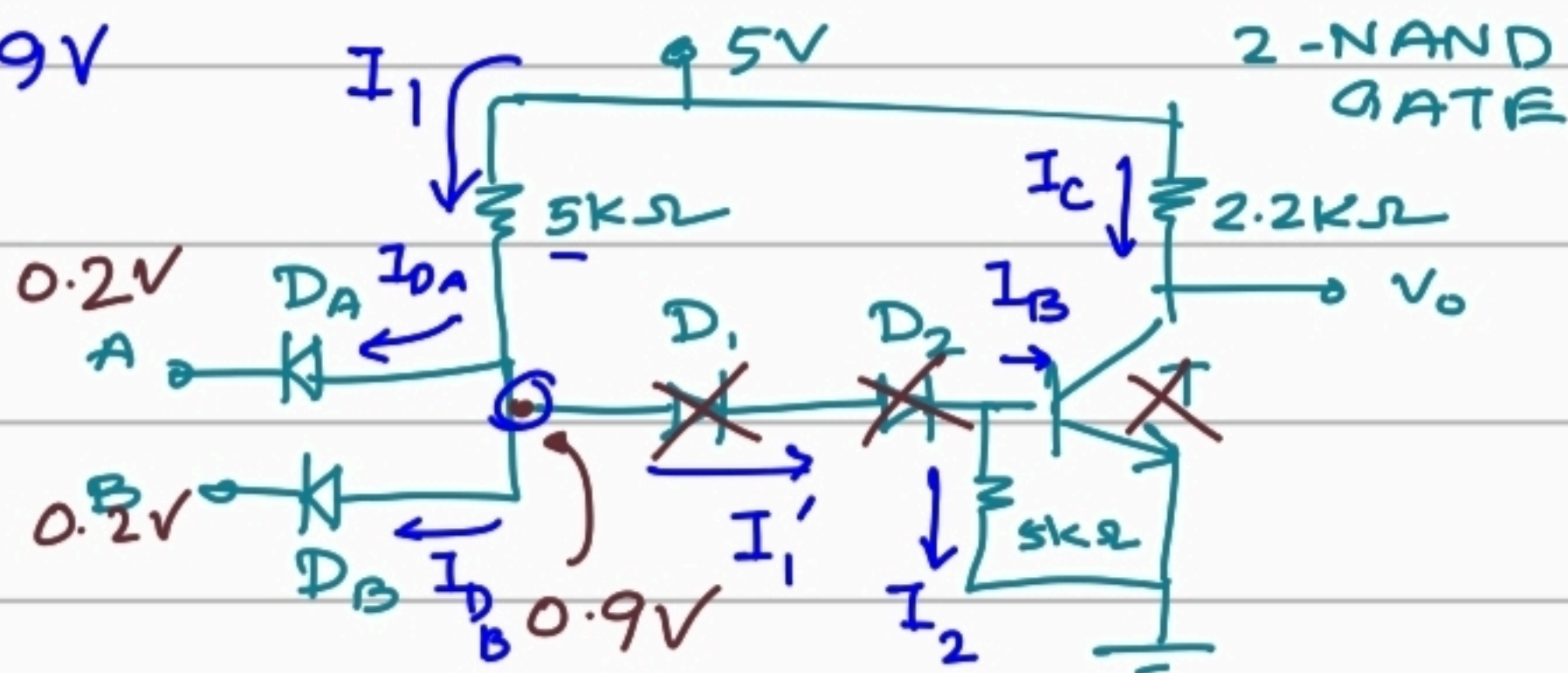
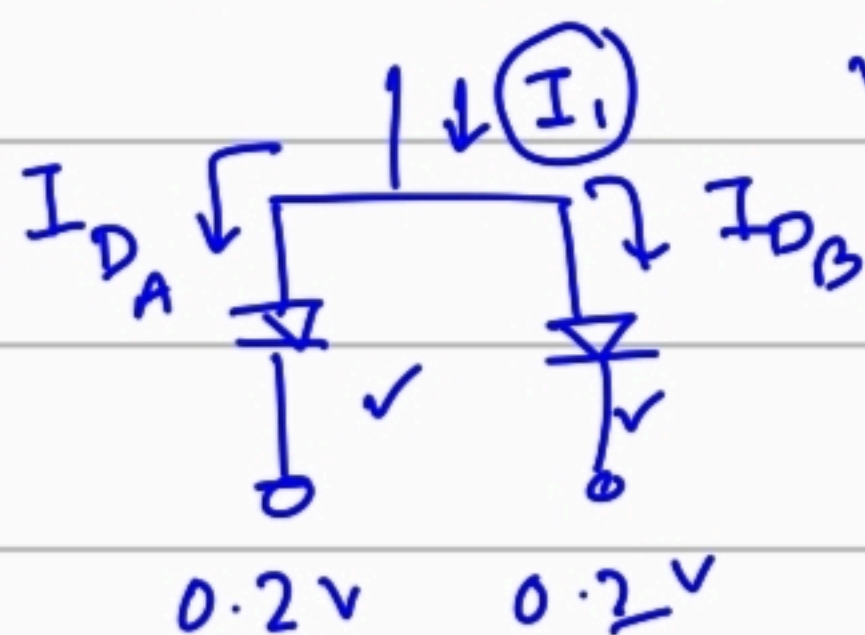
According to our assumption, V_P will not turn on D_1 , D_2 and T and current will flow as we expected.

$$V_A = V_B = 0.2V, V_P = 0.9V$$

$$I_1 = \frac{(5 - 0.9)}{5k} = 0.82mA$$

$$I_C = I_B = I_2 = I_1' = 0$$

$$V_0 = 5V$$



$I_{DA} = I_{DB}$ because both branches are identical.

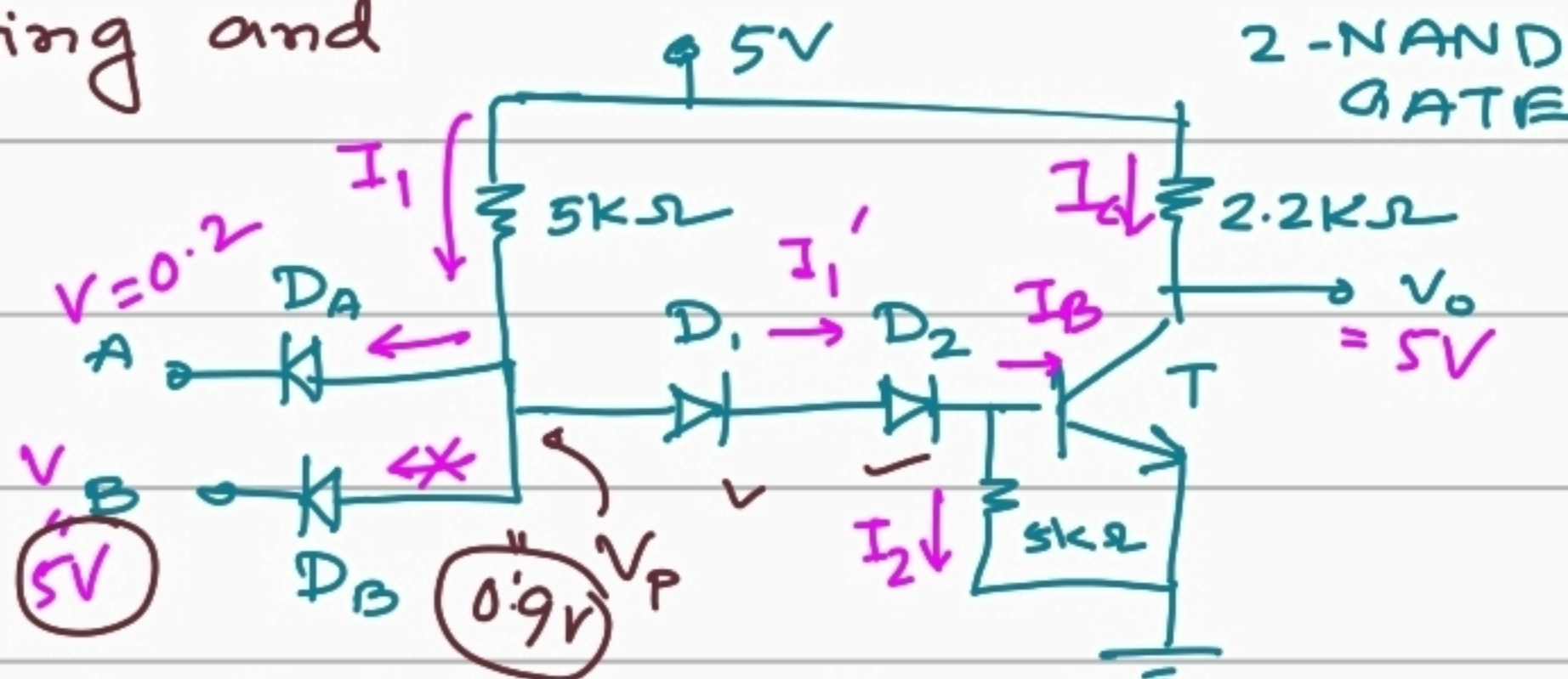
$$I_{DA} = I_{DB} = I_1/2 = 0.41mA$$

Case ② $V_A = 0.2V, V_B = 5V$ | ($V_A = 5V, V_B = 0.2V$)
Similar

Assume, D_A is conducting and D_B is off.

Therefore, $V_P = 0.9V$.

So D_1, D_2, T must be turned off.



The anode of D_B has smaller voltage than the cathode. So our assumption is right.

$$\# I_C = I_B = I_1' = I_2 = 0, V_O = 5V. I_1 = \frac{5 - 0.9}{5k\Omega} = 0.82 \text{ mA}$$

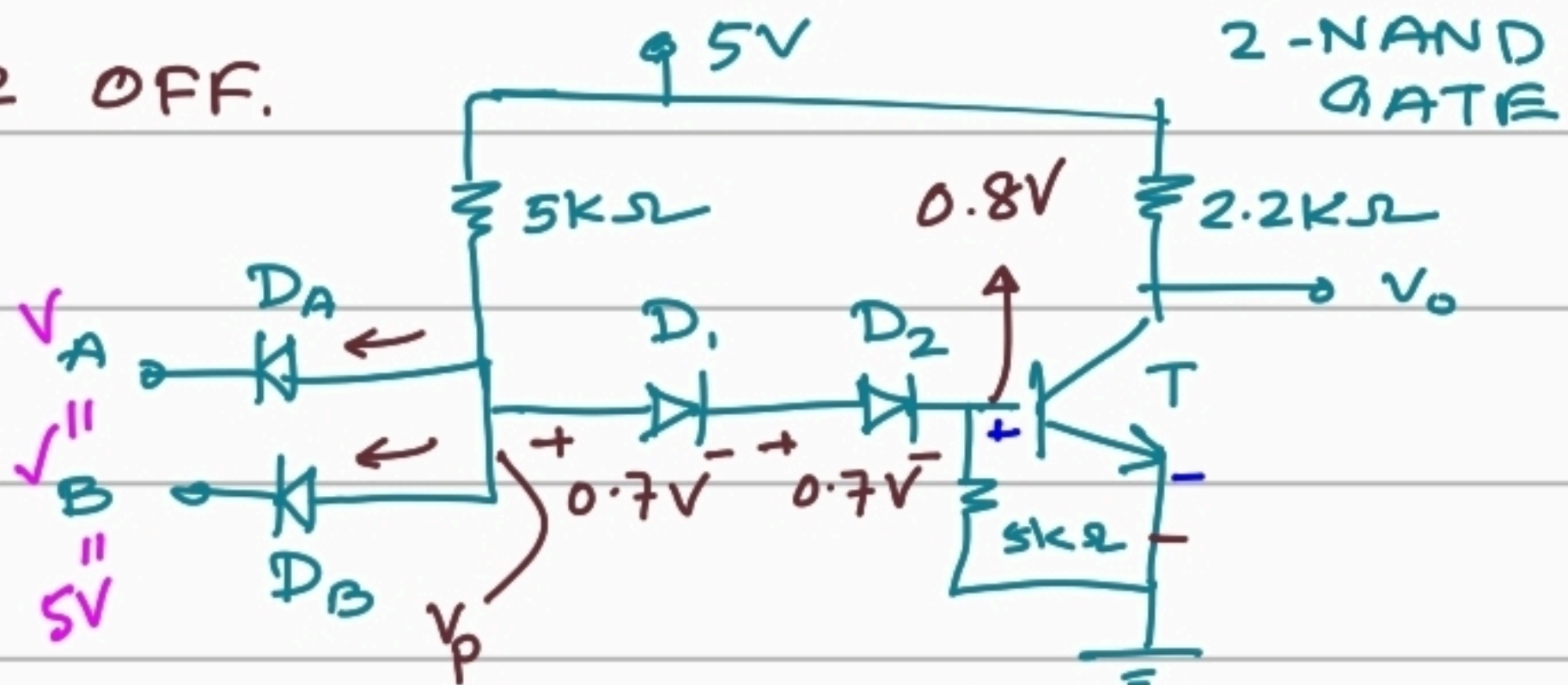
$$I_{D_A} = I_1 = 0.82 \text{ mA}, I_{D_B} = 0 \text{ mA}.$$

Case ③ $V_A = V_B = 5V$.

Assume, D_A and D_B are OFF.

$$I_{D_A} = I_{D_B} = 0 \text{ mA}.$$

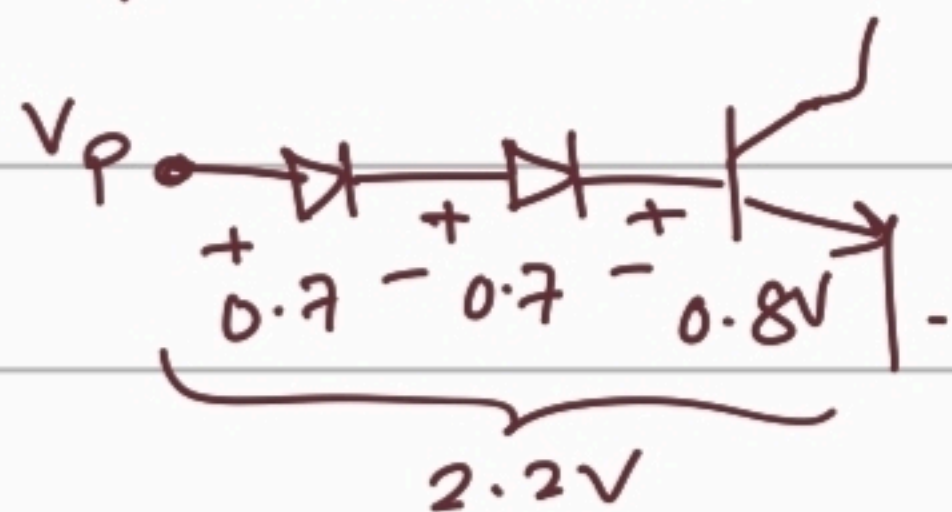
Therefore current must flow through D_1, D_2 and T .



Because it is a DTL circuit, the switching transistor must operate in saturation when it is turned on.

$$V_{BE}(\text{sat}) = 0.8V, V_{CE}(\text{sat}) = 0.2V$$

$\Rightarrow$ Conduction voltage of D_1 and D_2 diodes are $0.7V$.



According to our assumption

$$V_P \text{ must be } 2.2V.$$

The cathode voltages of D_A and D_B is higher than the anode voltage $V_P = 2.2V$

This justifies our assumption.

$\#$ Current-voltage Calculation:

$$V_p = 2.2V$$

$$I_1 = \left(\frac{5 - 2.2}{5k} \right) = 0.56mA$$

$$I_{D_A} = I_{D_B} = 0mA$$

$$I_1' = I_1 = 0.56mA$$

$$I_2 = \frac{0.8 - 0}{5k} = 0.16mA$$

$$I_B = I_1' - I_2 = 0.56 - 0.16 = 0.4mA$$

$$I_c = \frac{5 - 0.2}{2.2k} = 2.182mA$$

Chart:

V_A	V_B	V_O
0.2	0.2	5V
0.2	5V	5V
5V	0.2	5V
5	5	0.2

From analyzing the cases we can clearly conclude that this circuit is a NAND gate.

β_{min} for this circuit:

$$\beta_{forced} = \frac{I_c}{I_B} = \frac{2.182}{0.4} = 5.45$$

We need to choose $\beta_F \geq 5.45$ for transistor T in order to prevent it from malfunctioning.

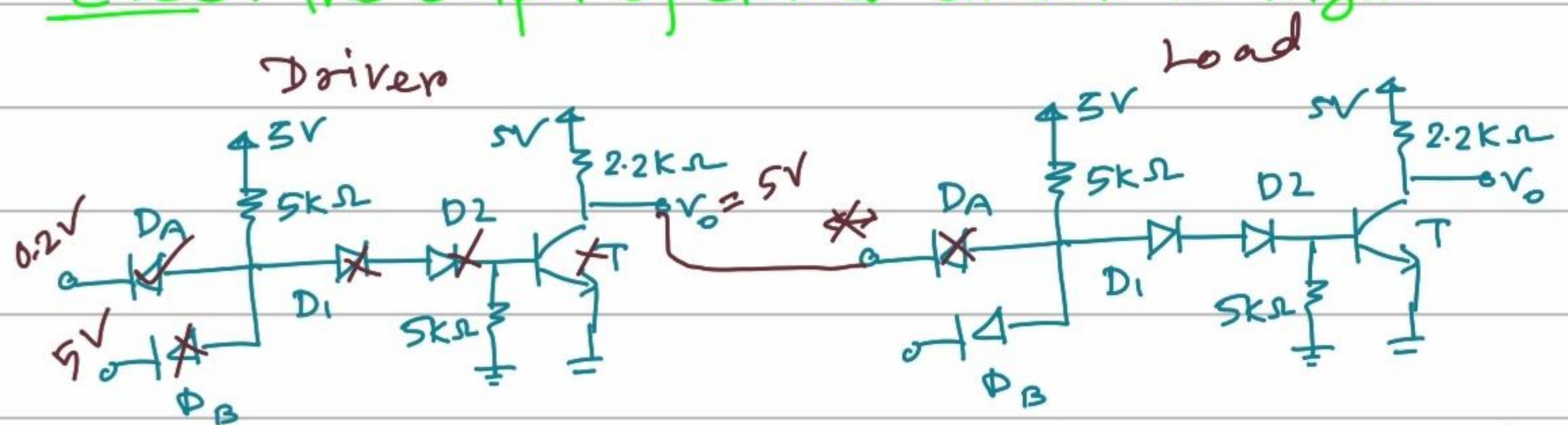
$$\beta_{min} = 5.45$$

Lecture 6: DTL

FANOUT,

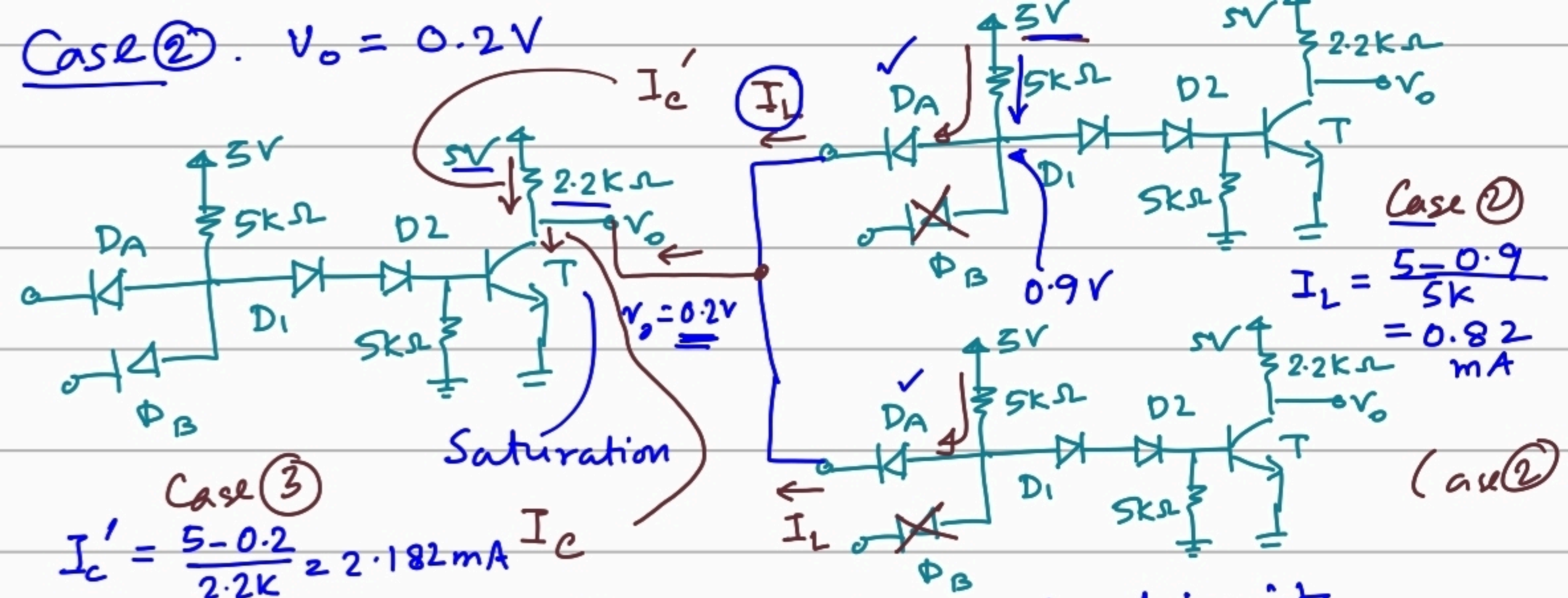
Maximum fanout Calculation :-

Case ①. The output of driver circuit is high.



∴ The demand current is zero, and supply is zero, also. Thus we can connect as much as load circuits we want. $\text{max Fanout} = \infty$.

Case ②. $V_o = 0.2V$



We want to consider worst

load circuits

case scenario, thus we will consider D_B of load circuits are off.

I_c = total collector current, I_c' = no. load collector current.

I_L = standard load (current), N = fanout

$$I_c = I_c' + N I_L = 2.182 + N \times 0.82$$

transistor

Constraint: When driver circuit's is operating in saturation mode, the collector current can not exceed a certain value. Otherwise, the driver transistor T will change its operating mode saturation to forward active.

Marginal situation will be achieved if

$\beta_{\text{forced}} \approx \beta_F$ for driver transistor T .

$$\beta_{\text{forced}} \approx \frac{I_c}{I_B} \Rightarrow I_{c,\text{max}} = \beta_F \times I_B$$

← Case ③

$$= 30 \times 0.4 = 1.2 \text{ mA}$$

Maximum fanout is N' . Then the main equation becomes,

$$I_{c,\text{max}} = I_c' + N' I_L$$

$$1.2 = 2.182 + N' \times 0.82$$

$$\Rightarrow N' = 11.96$$

We can not choose maximum fanout $N' = 12$

because that will exceed the value of permitted collector current.

Therefore, maximum fanout for this case (N')

$$\text{Finally maximum fanout} = \min(11, \infty) = 11$$